



Final Exam in
TDT4125 ALGORITHM CONSTRUCTION

Read the entire exam before you start!

It may be a good idea to tentatively start solving a difficult problem before setting it aside, keeping it on the back burner while working on something else.

Plan your time, and prepare any questions for when the teacher comes to the exam room.

Points are given for each subproblem. These sum to 120, but the maximum score is 100 points. In other words: **A score of more than 100 counts as 100.**

Problem 1

- a) Briefly explain how quantum parallelism works. Feel free to express your answer as an equation. (5 %)

There is a so-called *efficient* universal Turing machine U , which is such that for all $x, \alpha \in \{0, 1\}^*$, we have $U(x, \alpha) = M_\alpha(x)$, where M_α is the Turing machine represented by α . Assume that M_α with input x halts after T steps.

- b) What can you say about how many steps U uses with inputs x and α ? (5 %)
- c) You have a linear program where at least one of two given constraints must hold. You may assume that the variables are non-negative and have an upper limit, that is, $0 \leq x_j \leq u_j$ for every j . Explain how you would handle this with integer programming. (10 %)

- d) Explain briefly what $T_P(n)$ is, and give two lower bounds for it. Explain why these bounds hold. No computations are needed; an intuitive explanation suffices, preferably linked to the computation DAG. (10 %)
- e) Briefly explain how the Euler tour technique is used to find the depth of all n nodes of a binary tree. What is the running time? (10 %)

Problem 2

- a) Explain the definitions of the classes P, NP, co-NP, NP-hard og NPC. (10 %)

Your two co-students Lurvik and Smartnes are haveing a heated discussion, where Lurvik claims that 2-CNF satisfiability is NP complete, while Smartnes claims that it is not.

- b) What is your opinion in the matter? Explain your answer. (10 %)

Two graphs $G = (V, E)$ and $G' = (V', E')$ are *isomorphic* if there is a bijection (one-to-one correspondence) $f : V \rightarrow V'$ such that $(u, v) \in E$ if and only if $(f(u), f(v)) \in E'$. Let $\text{GRAPH-ISOMORPHISM} = \{\langle G, G' \rangle : G \text{ og } G' \text{ are isomorphic graphs}\}$.

- c) Show that $\text{GRAPH-ISOMORPHISM} \in \text{NP}$. (10 %)
- d) Assume that you have an algorithm A that finds the shortest simple path between two nodes s and t in a complete, weighted, undirected graph. Assume that A has a running time of $T(n)$, where n is the number of nodes. Show how A may be used to solve the *Hamilton cycle* problem (HAM-CYCLE) with a running time of $T(n) \cdot O(n^k)$, for some k . Try to make k as low as possible. Discuss what k is for your solution. (10 %)

You wish to hold a party for friends and acquaintances, but don't want the hassle of inviting everyone. Instead, you want to select as few as possible that combined know everyone, and get these to spread the invitation to the others. The book *Algorithms to Live By* av Christian & Griffiths (2016) suggests a rounded continuous relaxation of this problem, that is, that you find an optimal solution to the problem where everyone may receive a fractional invitation, and then round these off to 0 or 1. The authors then say:

It turns out that for the invitations problem, continuous relaxation with rounding will give us an easily computed solution that's not half bad: it's mathematically guaranteed to get everyone you want to the party while sending out at most twice as many invitations as the best solution obtainable by brute force. (s. 177)

Without knowing exactly which constraints are used for the fractional invitations, we cannot know exactly how they reached this conclusion. Even so, it is possible to draw some general conclusions.

- e) Discuss the claims of Christian & Griffiths. Which problem in the curriculum does the invitation problem correspond to? Explain your answer relatively thoroughly. What can you say about approximation of this problem? Do you know of any similar problems that fits their reasoning? (10 %)

Problem 3

- a) Explain what a parameterized problem is, and what it means for a problem to be fixed-parameter tractable. (10 %)
- b) When you want to find an edge clique cover and two nodes u and v have the same closed neighborhood (the set of neighbors and the node itself), why is it safe to reduce the problem by deleting one of the nodes? (10 %)

A *matching* in an undirected graph $G = (V, E)$ is a set of disjoint edges, that is, a subset M of E where no edges in M share nodes. A *maximal* matching is one that cannot be extended with edges from E and still be a matching. You are given an undirected graph G and a positive integer k , and wish to decide whether there is a maximal matching in G with at most k edges.

- c) Describe a polynomial kernel for the problem, parameterized by k . Explain your approach and show your calculations. (You may describe your approach in your own words; you need not use a list of reduction rules, as in the textbook.) (10 %)

Hint: What can you say about the end points of the matching?